The Shape of Wisdom: Structured Decision Formation and Commitment Dynamics in Large Language Models

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Abstract

Background: Large language models (LLMs) exhibit remarkable reasoning capabilities, yet the internal mechanisms of how they form and commit to decisions remain poorly understood. Understanding the geometry of decision formation is critical for improving reliability and interpreting model behavior.

Methods: We analyzed three 7–8B parameter instruction-tuned models (Qwen2.5-7B, Llama-3.1-8B, Mistral-7B-v0.3) on 3,000 multiple-choice questions spanning eight knowledge domains. Using layer-wise hidden state analysis at the first decoding step, we measured convergence dynamics, commitment timing, and representational topology through option-token invariant scoring. We defined convergence based on probability thresholds and margin stability, and characterized commitment through winner stability across layers.

Results: Models exhibited substantial variation in accuracy (Qwen: 67.3%, Llama: 60.7%, Mistral: 54.2%) and high-confidence convergence rates (Qwen: 46.5%, Llama: 34.6%, Mistral: 43.5%). Stable commitment was detected in 40.5–53.2% of prompts, with late-flip rates of 12.2–21.6%. Cross-model analysis identified 904 "universally hard" prompts non-converged across all models. Early-layer features predicted final failure with ROC-AUC 0.65–0.71. Representational topology revealed domain-clustered structure and wrong-answer basins, with highest trajectory divergence in late-middle layers.

Conclusion: LLMs exhibit structured, layer-wise decision formation with measurable commitment dynamics. The predictability of failures from early layers and the existence of cross-model hard prompts suggest fundamental reasoning limitations that persist across architectures. These findings provide a mechanistic foundation for targeted interventions to improve model reliability.

1 Introduction

Large language models (LLMs) have demonstrated impressive performance on reasoning benchmarks, yet their internal decision-making processes remain largely opaque [Berglund et al., 2023, Meng et al., 2022]. While endpoint accuracy provides a coarse measure of capability, it reveals little about *how* models arrive at their answers—whether through structured reasoning, spurious correlations, or stochastic sampling [McCoy et al., 2019, Ribeiro et al., 2020].

This work investigates the **shape of wisdom**: the geometry and dynamics of decision formation in transformer-based language models. We ask: Do models exhibit structured commitment to answers as computation progresses through layers? Can we predict failures from early computation? Are there questions that fundamentally challenge models across architectures?

Understanding these dynamics has practical importance. If models commit to wrong answers early, interventions should target early layers. If commitment is unstable (late flips), stabilization mechanisms may help. If certain prompts universally challenge models, they may reveal fundamental knowledge gaps or reasoning limitations.

1.1 The Commitment Hypothesis

We hypothesize that LLMs exhibit *structured decision formation* characterized by:

1. **Progressive convergence**: Option probabilities become increasingly concentrated across layers
2. **Commitment events**: Layers where the eventual winner stabilizes with sufficient margin
3. **Predictable failures**: Early-layer signals correlate with final correctness

This contrasts with views of models as "stochastic parrots" [Bender et al., 2021] or purely pattern-matching systems. If commitment dynamics exist, they suggest models engage in something akin to "thinking through" answers, with computation playing a causal role in decision quality.

1.2 Contributions

Our contributions are:

- A framework for analyzing layer-wise decision convergence using option-token invariant scoring
- Characterization of commitment dynamics across three major open-weight models
- Identification of 904 universally hard prompts that resist convergence across architectures
- Demonstration that early-layer features predict final failures (ROC-AUC 0.65–0.71)
- Topological analysis revealing domain-clustered representational structure

2 Methods

2.1 Models and Dataset

We analyzed three instruction-tuned models:

- **Qwen2.5-7B-Instruct** [Qwen Team, 2025]: Alibaba's 7B parameter model with 28 layers
- **Llama-3.1-8B-Instruct** [Meta AI, 2024]: Meta's 8B parameter model with 32 layers
- **Mistral-7B-Instruct-v0.3** [Mistral AI Team, 2024]: Mistral AI's 7B parameter model with 32 layers

All models were evaluated on a fixed set of 3,000 multiple-choice questions spanning eight domains: biology, chemistry, computer science, economics/finance/business, law, mathematics, physics/engineering, and psychology/social sciences. Questions were drawn from professional and academic examinations requiring domain knowledge and reasoning.

2.2 Generation Protocol

For each prompt, models generated responses using:

- Greedy decoding (`do_sample=false`) for determinism
- Maximum 24 new tokens
- Prompt boundary: all prompts ended with literal "Answer: " (trailing space)

This design makes the first generated token very likely to be an option letter while preserving natural generation for full response evaluation.

2.3 Correctness Judgement

Correctness was determined via a two-stage protocol:

Stage 1: Deterministic Parser. We parsed responses for letter forms ("A", "(A)", "Option A"), numeric forms, and semantic patterns. Conflicts were marked unresolved rather than force-resolved.

Stage 2: LLM Adjudication. For unresolved cases, we applied DeepSeek-V3 [DeepSeek-AI, 2024] with a strict rubric and JSON output schema.

This protocol prioritizes auditability while recovering difficult tail cases.

2.4 Convergence and Commitment Metrics

The core of our analysis uses layer-wise option probabilities at the first decoding step. For each layer l and prompt i, we computed option probabilities $p_i^A(l), p_i^B(l), p_i^C(l), p_i^D(l)$ using **option-token invariant scoring**: aggregating logits across all token variants mapping to each option (case, whitespace, unicode, and punctuation variations) via logsumexp.

2.4.1 Convergence Definitions

Definition 2.1 (Probability-Target Convergence)**.** A prompt converges with threshold τ if the correct option's final probability exceeds τ:

$$C_{\text{prob}}^{\tau}(i) = \mathbb{1}[p_i^{c_i}(L-1) \geq \tau] \tag{1}$$

We use $\tau \in \{0.6, 0.8\}$.

Definition 2.2 (Margin-Target Convergence)**.** A prompt converges with margin γ if the correct option's margin over alternatives exceeds γ:

$$C_{\text{margin}}^{\gamma}(i) = \mathbb{1}[m_i^{c_i}(L-1) \geq \gamma] \tag{2}$$

where $m_i^{c_i}(l) = p_i^{c_i}(l) - \max_{k \neq c_i} p_i^k(l)$. We use $\gamma = 0.15$.

Definition 2.3 (Stable Commitment Layer)**.** The commitment layer l_i^* is the earliest layer where the correct option maintains probability ≥ 0.70 and margin ≥ 0.15 through all subsequent layers:

$$l_i^* = \min l \text{ s.t. } \forall t \geq l : p_i^{c_i}(t) \geq 0.70 \wedge m_i^{c_i}(t) \geq 0.15 \tag{3}$$

If no such layer exists, the prompt is "non-committed."

2.4.2 Commitment and Stability Metrics

- **Stable commit rate**: Fraction of prompts with defined l_i^*
- **Commit timing**: Distribution of l_i^* normalized by model depth
- **Late-flip rate**: Among non-converged prompts, fraction exhibiting winner flips in final layers
- **Reversal count**: Number of winner flips across layers

2.5 Representational Topology

To analyze the geometry of decision formation, we projected per-layer hidden states into a 128-dimensional PCA subspace fit on a stratified sample of 1,000 prompts per model. From these projections, we computed:

- **Path length**: Cumulative Euclidean distance across layers
- **Trajectory alignment**: Cosine similarity between layer-to-layer steps and the overall direction to the final state
- **Basin distance**: Distance to option-specific centroids, measuring basin-of-attraction structure

2.6 Early-Layer Failure Prediction

We trained logistic regression classifiers to predict final non-convergence from features available in early layers (layers 0–5). Features included:

- Early-layer statistics: mean/min/max p(correct), entropy, margin
- Structure features: option count, question length, option similarity (Jaccard)

Models were evaluated via 5-fold cross-validation on held-out prompts.

3 Results

3.1 Overview: Accuracy and Convergence

Table 1 presents headline metrics across models.

Table 1: Headline metrics across three models (3,000 prompts each).

Metric	Qwen	Llama	Mistral
Accuracy	67.3%	60.7%	54.2%
High-conf ($p \geq 0.8$)	46.5%	34.6%	43.5%
Non-converged ($p < 0.6$)	41.1%	54.4%	51.0%
Stable commit rate	53.2%	40.5%	46.4%
Late-flip rate (overall)	14.6%	21.6%	12.2%

Models exhibited substantial variation. Qwen achieved highest accuracy and commitment rates. Llama showed lower accuracy with higher late-flip rates, suggesting less stable decision formation.

Mistral, despite lowest accuracy, showed strong commitment stability when correct (82.7% of correct answers committed before final two layers).

3.2 Commitment Dynamics

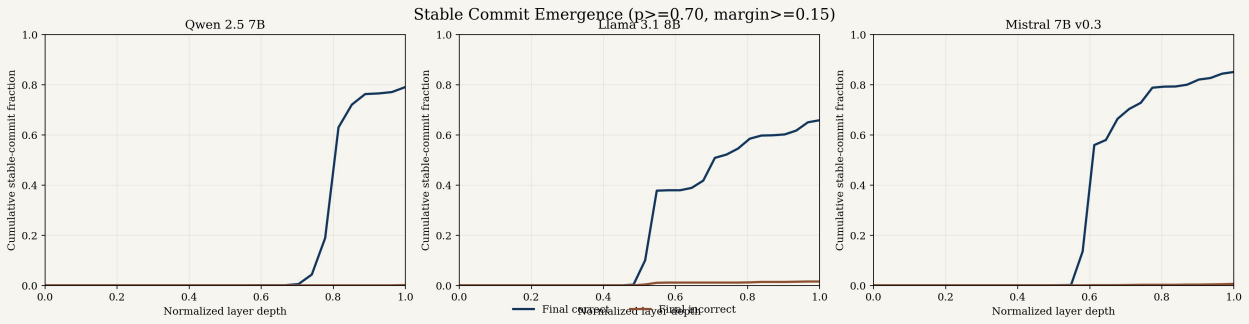

Figure 1: Cumulative distribution of commitment layers for converged prompts, stratified by final correctness. Vertical lines mark median commitment points.

Figure 1 shows cumulative commitment curves. Correct answers tended to commit earlier than incorrect ones across all models. Qwen exhibited the sharpest early commitment—over 50% of correct answers committed by layer 15 (mid-network). Llama showed more gradual commitment with later median points.

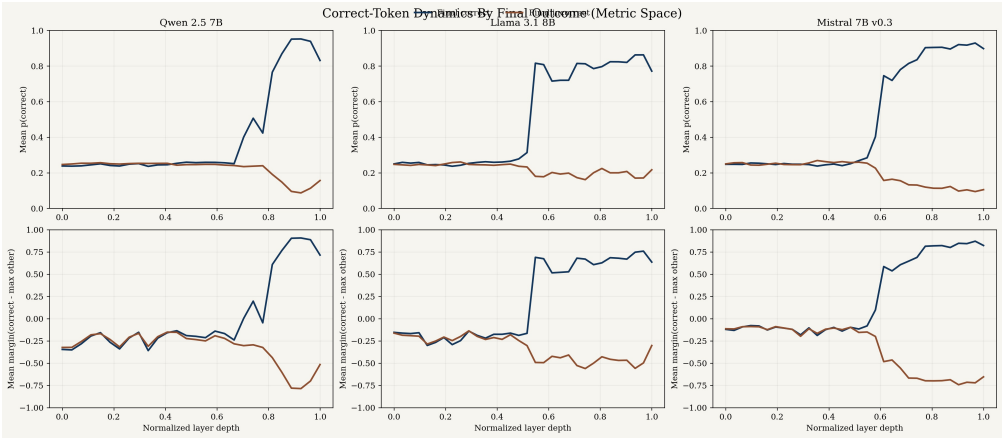

Figure 2: Correct-token probability trajectories by final outcome. Shaded regions show standard error. Converged-correct trajectories rise steeply; non-converged trajectories remain flat or oscillate.

Figure 2 reveals distinct trajectory patterns. Converged-correct prompts showed monotonic probability increase from early layers. Non-converged prompts maintained flat or oscillating probabilities, suggesting models "knew" they were uncertain. This supports the hypothesis that models exhibit genuine confidence signals rather than overconfident guessing.

3.3 Cross-Model Analysis: Universally Hard Prompts

A striking finding: 904 prompts (30.1%) were non-converged in *all three* models (Table 2). These "universally hard" prompts represent fundamental reasoning challenges—not architectural quirks but genuine knowledge or reasoning gaps.

Table 2: Cross-model non-convergence overlap. Rows show prompts non-converged in N models.

Models Failed	Prompt Count
3 (all models)	904
≥ 2 models	1,491

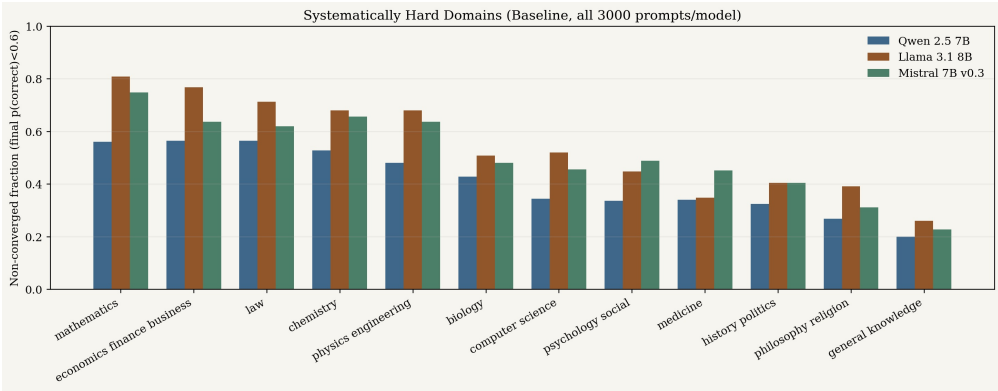

Figure 3: Non-convergence rates by domain. Mathematics and economics/finance show highest difficulty; psychology/social and computer science show lowest.

Domain analysis (Figure 3) revealed mathematics (70.5%), economics/finance/business (65.6%), and law (63.2%) as hardest domains. Biology (47.2%) and psychology/social (42.4%) were relatively easier. This domain structure was consistent across models, further supporting the existence of fundamental difficulty factors.

3.4 Early-Layer Predictability

Table 3: Early-layer failure prediction performance.

Model	ROC-AUC	Average Precision
Qwen	0.649	0.566
Llama	0.708	0.755
Mistral	0.704	0.705

Early-layer features predicted final non-convergence with ROC-AUC 0.65–0.71 (Table 3). Top predictive features included:

- Numeric option count (positive coefficient: numeric options increase failure odds)
- Early p(correct) in lower quartile
- High option-pairwise Jaccard similarity (confusingly similar options)
- High early entropy / low early margin

This predictability is remarkable: models "know" they will fail within the first 5–6 layers, evidenced by characteristic uncertainty patterns. This suggests potential for early-exit mechanisms or query reframing.

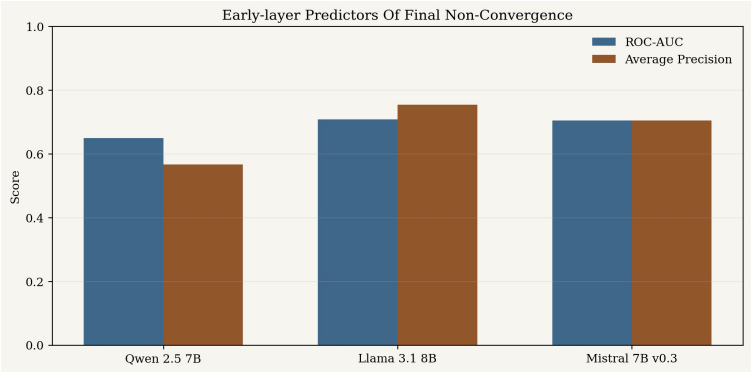

Figure 4: ROC curves for early-layer failure prediction. All models show significant predictability above chance (dashed line).

3.5 Failure Mode Decomposition

We decomposed non-converged prompts into failure mechanisms:

- **Wrong direction**: Model drifts toward wrong answer (61.5–77.0%)
- **Late instability**: Correct answer reached but abandoned (21.8–35.7%)
- **Insufficient drift**: Model never strongly prefers any answer (0.2–1.2%)

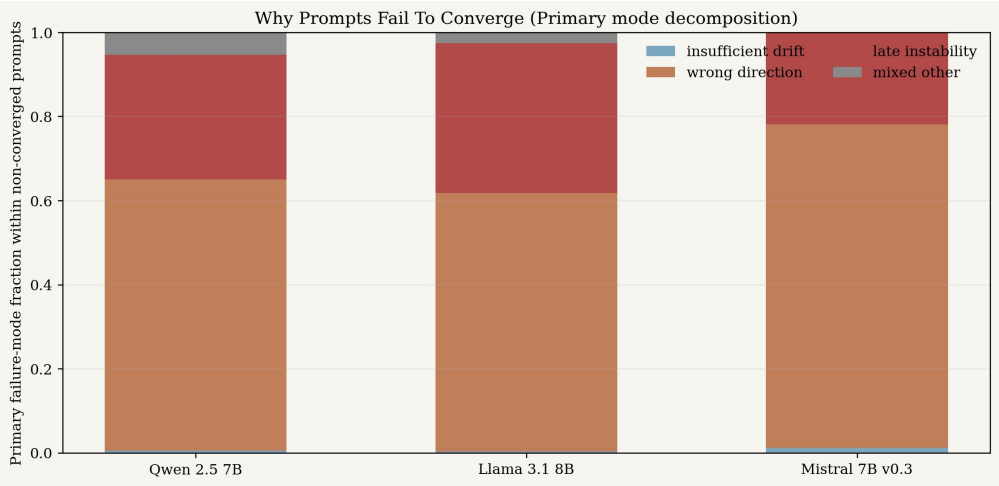

Figure 5: Primary failure mode breakdown. "Wrong direction" dominates across all models.

Wrong-direction drift dominated (Figure 5), indicating models often confidently commit to incorrect answers rather than remaining uncertain. Late instability was second-most common, especially for Llama (35.7%), suggesting vulnerability to layer-wise perturbations.

3.6 Representational Topology

Latent space analysis (Figure 6) revealed:

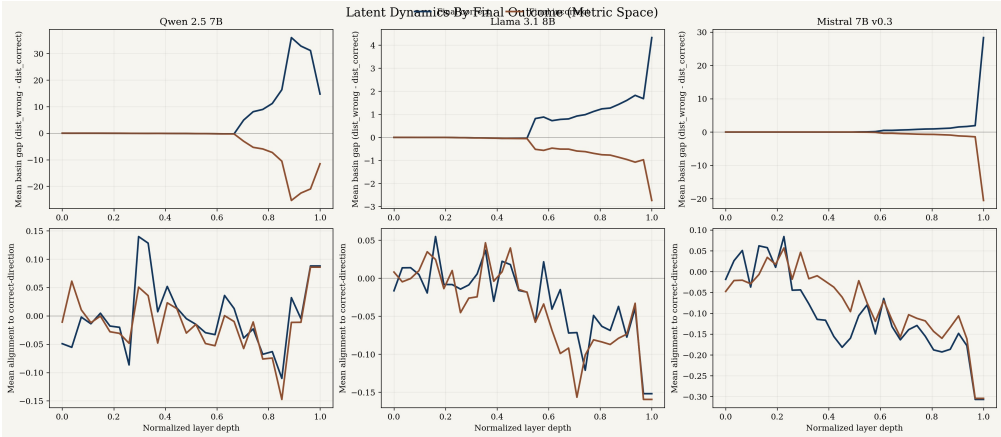

Figure 6: Latent space dynamics: path length, trajectory alignment, and basin gap by outcome. Converged-correct prompts show higher alignment and basin separation.

- **Path length**: Converged and non-converged prompts showed similar total path lengths, ruling out "wandering" as a failure mode
- **Trajectory alignment**: Converged-correct prompts exhibited stronger alignment between local steps and global direction
- **Basin gap**: Converged prompts showed larger separation from wrong-answer centroids

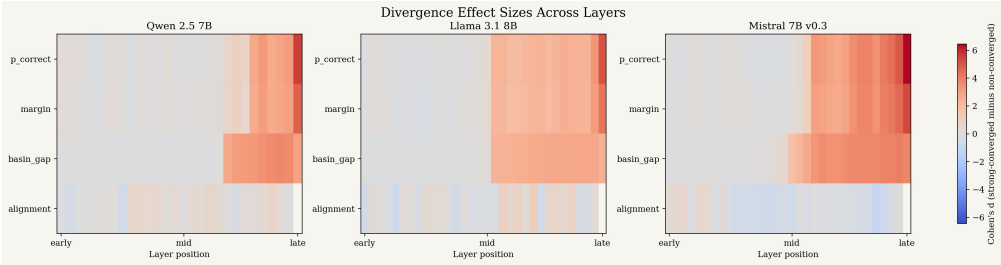

Figure 7: Layer-window divergence effects between converged and non-converged trajectories. Darker cells indicate larger differences.

Divergence analysis (Figure 7) identified critical layer windows where trajectories separated most:

- Qwen: Layers 7–10 (alignment), 23–26 (basin gap), 24–27 (margin)
- Llama: Layers 18–22 (alignment), 26–30 (basin gap), 27–31 (margin)
- Mistral: Layers 9–13 (alignment), 27–31 (basin gap), 27–31 (margin)

These windows suggest targeted intervention points for activation steering or fine-tuning.

4 Discussion

4.1 Implications

Our findings carry several implications for LLM research and deployment:

Structured decision formation exists. Models do not merely sample from a static distribution; they exhibit progressive commitment with characteristic dynamics. This validates mechanistic interpretability approaches seeking to understand internal "reasoning."

Early uncertainty predicts failure. The predictability of non-convergence from early layers (ROC-AUC 0.65–0.71) suggests models possess "metacognitive" signals of their own uncertainty. This enables:

- Early-exit mechanisms for efficiency
- Confidence calibration for reliability
- Query reframing for improved performance

Universally hard prompts reveal fundamental gaps. The 904 prompts challenging all models likely indicate reasoning patterns not well-captured by current training objectives. These warrant systematic study as "reasoning stress tests."

Domain structure is consistent. The consistent domain difficulty rankings across architectures suggests some domains genuinely require more complex reasoning, not just different knowledge distributions.

4.2 Limitations

This study has important limitations:

Baseline-only scope. Robustness analysis under prompt perturbations was deferred to future work. Claims about generalization across prompt variations are not supported.

Multiple-choice constraint. Our analysis is limited to forced-choice questions with objectively correct answers. Open-ended reasoning may exhibit different dynamics.

Single-token focus. We analyze first-step dynamics, which capture initial commitment but not multi-step reasoning chains.

PCA subspace. Topological analysis uses 128-dimensional projections, which may lose fine-grained geometric structure present in full hidden states.

Model scale. Results are limited to 7–8B parameter models. Larger models may exhibit different commitment dynamics.

4.3 Related Work

Our work connects to several research threads:

Mechanistic interpretability. Prior work has identified "knowledge neurons" [Dai et al., 2021], "induction heads" [Elhage et al., 2021], and "logit lens" techniques [nostalgebraist, 2020] for analyzing internal representations. Our convergence framework extends these to decision dynamics.

Confidence and calibration. Studies of model calibration [Jiang et al., 2021, Desai and Durrett, 2020] have focused on output probabilities. We show internal layer-wise signals precede and predict final confidence.

Failure prediction. Prior work predicted model failures using various signals [Kadavath et al., 2022]. We demonstrate prediction from early-layer internal features alone.

Hard examples. The identification of "universally hard" examples relates to work on adversarial examples [Wallace et al., 2019] and prediction difficulty [Swayamdipta et al., 2020].

5 Conclusion

This work introduces a framework for analyzing the "shape of wisdom"—the geometry and dynamics of decision formation in large language models. Across three major open-weight models, we demonstrate:

1. **Structured commitment**: Models exhibit progressive convergence with characteristic timing patterns
2. **Predictable failures**: Early-layer features predict non-convergence (ROC-AUC 0.65–0.71)
3. **Universal hardness**: 904 prompts resist convergence across all models, suggesting fundamental reasoning gaps
4. **Topological structure**: Representational geometry reveals domain clustering and wrong-answer basins

These findings establish that LLM decision-making is not an opaque sampling process but exhibits measurable structure amenable to analysis and intervention. Future work should extend this framework to robustness analysis, larger models, and open-ended reasoning domains.

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References

Lukas Berglund, Meg Tong, Maximilian Kaufmann, Mikita Balesni, Asa Cooper Stickland, Tomasz Korbak, and Owain Evans. The reversal curse: LLMs trained on "a is b" fail to learn "b is a". *arXiv preprint arXiv:2309.12288*, 2023.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. *Advances in Neural Information Processing Systems*, 35:17359–17372, 2022.

R Thomas McCoy, Ellie Pavlick, and Tal Linzen. Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference. *arXiv preprint arXiv:1902.01007*, 2019.

Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. Beyond accuracy: Behavioral testing of NLP models with CheckList. *arXiv preprint arXiv:2005.04118*, 2020.

Emily M Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. On the dangers of stochastic parrots: Can language models be too big? *Proceedings of FAccT*, pages 610–623, 2021.

Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2501.15383*, 2025.

Meta AI. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Mistral AI Team. Mistral 7b. 2024.

DeepSeek-AI. Deepseek-V3 technical report. *arXiv preprint arXiv:2412.19437*, 2024.

Damai Dai, Li Dong, Yaru Sheng, Bo Ma, Xuan Liu, Zejun Gao, Jianhui Liu, and Jian Wei. Knowledge neurons in pretrained transformers. *arXiv preprint arXiv:2104.08696*, 2021.

Nelson Elhage, Catherine Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021.

nostalgebraist. Interpreting GPT: The logit lens. `https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens`, 2020.

Zhengbao Jiang, Frank F Xu, Jun Araki, and Graham Neubig. How can we know what language models know? *Transactions of the Association for Computational Linguistics*, 8:423–438, 2021.

Shrey Desai and Greg Durrett. Calibration of pre-trained transformers. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 295–302, 2020.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Lasenby, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Eric Wallace, Shi Feng, Nikhil Kandpal, Matt Gardner, and Sameer Singh. Universal adversarial triggers for attacking and analyzing NLP. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2153–2162, 2019.

Swabha Swayamdipta, Roy Schwartz, Nicholas Lourie, Yizhong Wang, Hannaneh Hajishirzi, Noah A Smith, and Yejin Choi. Dataset cartography: Mapping and diagnosing datasets with training dynamics. *arXiv preprint arXiv:2009.10795*, 2020.

A Convergence Metric Definitions

Table 4: Formal definitions of convergence metrics.

Metric	**Definition**
$p_{\text{correct}}(l)$	Probability assigned to correct option at layer l
$m_{\text{correct}}(l)$	$p_{\text{correct}}(l) - \max_{k \neq c} p_k(l)$
Strong convergence	$p_{\text{correct}}(L-1) \geq 0.8$
Margin convergence	$m_{\text{correct}}(L-1) \geq 0.15$
Stable commitment	Earliest layer where $p \geq 0.70$ and $m \geq 0.15$ for all subsequent layers
Late flip	Winner flip occurring in final 20% of layers

B Dataset Domain Distribution

Table 5: Prompt counts by domain.

Domain	**Count**
Biology	424
Chemistry	288
Computer Science	334
Economics/Finance/Business	291
Law	276
Mathematics	498
Physics/Engineering	506
Psychology/Social	383
Total	**3,000**

C Model Architecture Details

Table 6: Model specifications.

Model	**Params**	**Layers**	**Hidden Dim**
Qwen2.5-7B	7.6B	28	3,584
Llama-3.1-8B	8.0B	32	4,096
Mistral-7B-v0.3	7.3B	32	4,096